AIORI HACKATHON 3 · PS-032

Path & Reachability Analysis During RPKI State Changes

TEAM DEBUGS

A precise, event-driven view of how routing validity changes propagate into reachability risk.

AIORI HACKATHON 3

Meet Team Debugs

Three minds. One mission.

JASMEET

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Team Lead
problem understanding and solution design

NISHANT

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Data Engineer
analysis pipeline

UMA

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Results Analyst
case studies, report and demo

Problem Description

RPKI state is not static. ROA additions, withdrawals, and edits can instantly change route validity — but operators still lack a fast explanation of which AS paths and destinations are newly exposed.

THE GAP

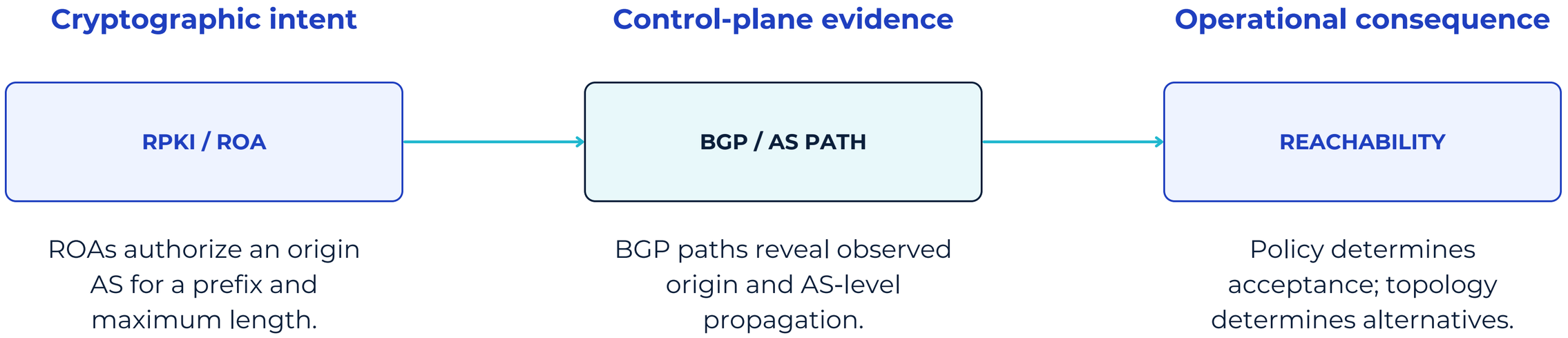
- Validation feeds show state, not impact
- Full recomputation is slow at Internet scale
- Invalid does not always mean unreachable
- Root cause is buried across time and topology

Required outcome

Given an RPKI state change, identify affected prefixes, changed paths, reachability consequences, and the evidence behind each result.

Background

Three layers determine whether a route is trusted and usable.



Key distinction: ROV validity is a policy input — not a direct reachability verdict.

Literature and RFC Review

[1] RFC 6480

Defines the RPKI architecture and trust model.

[2] RFC 6811 / 8893

Specifies origin validation and clarifies operational use.

[3] RFC 8205

BGPsec path validation: relevant context, distinct from ROV.

[4] RFC 8416

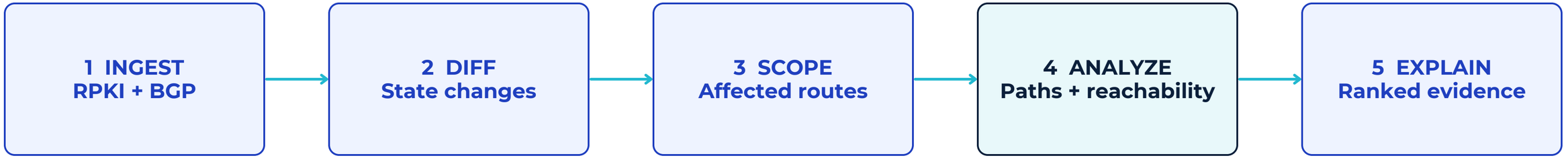
Operational guidance for route origin validation deployment.

Design implication

Keep cryptographic validity, observed path change, and inferred reachability as separate evidence layers.

Proposed Solution

An incremental, evidence-preserving five-step pipeline



Normalize snapshots and event streams

Detect ROA add, withdraw, modify

Map prefixes, origins, paths and peers

Recompute only impacted graph regions

Emit severity, cause and trace

OUTPUT · affected prefix → changed validity → path delta → reachability confidence → evidence

Optimization Proposed

Move from periodic global recomputation to targeted incremental analysis.

DIMENSION	BEFORE · NAIVE	AFTER · PROPOSED
Trigger	Periodic full scan	Event-driven state diff
Scope	All prefixes and paths	Impacted prefix / origin subgraph
Computation	Repeated graph rebuild	Cached indices + incremental update
Output	Validity snapshot	Path delta + reachability evidence
Operator value	Slow, noisy triage	Ranked, explainable alerts

Solution Architecture and Design

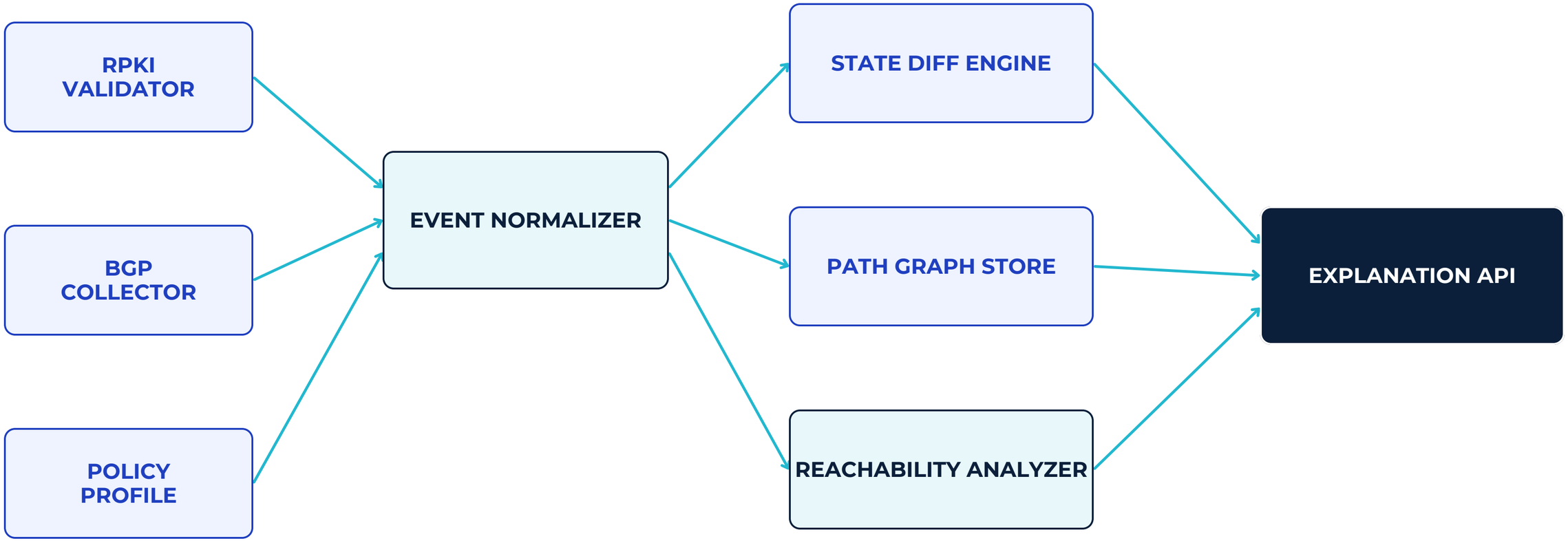


Figure: System Architecture UML Diagram

Data Flow Diagram

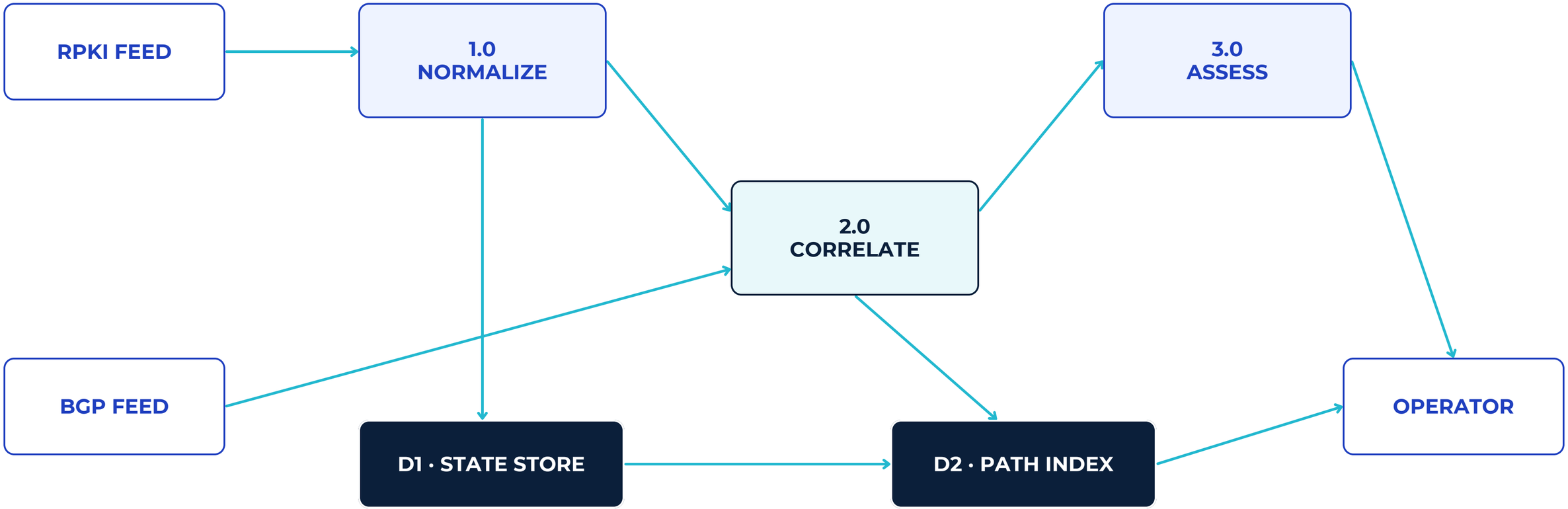
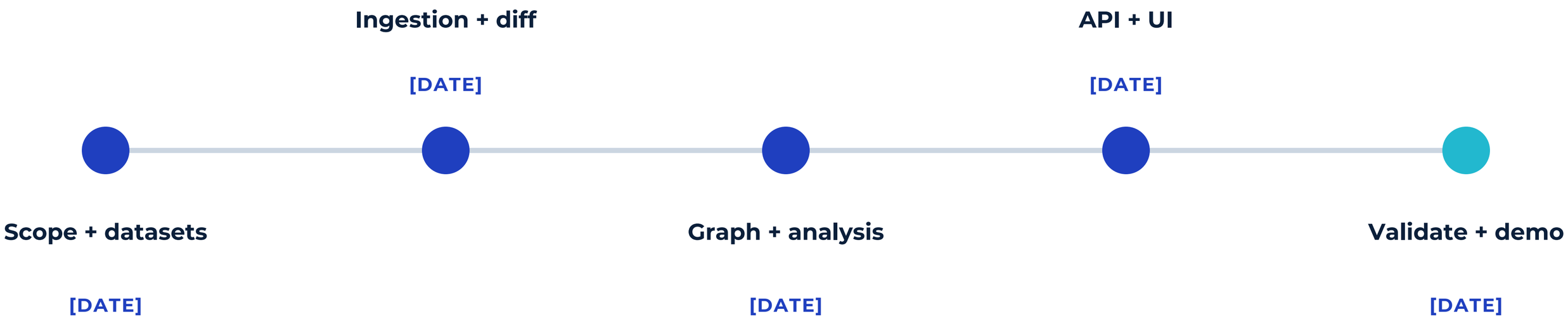


Figure: Data Flow Diagram (DFD)

Deployment Timeline

Milestones remain editable; replace date placeholders after sprint planning.



Bibliography

[1] Lepinski, M.; Kent, S. RFC 6480: An Infrastructure to Support Secure Internet Routing. IETF, 2012.

[2] Mohapatra, P. et al. RFC 6811: BGP Prefix Origin Validation; updated operational guidance in RFC 8893. IETF, 2013 / 2020.

[3] Lepinski, M.; Sriram, K. RFC 8205: BGPsec Protocol Specification. IETF, 2017.

[4] Bush, R. RFC 8416: Simplified Local Internet Number Resource Management with the RPKI. IETF, 2018.

[5] Gilad, Y. et al. MaxLength Considered Harmful to the RPKI. CoNEXT, 2017.

Thank You for Your Attention!

QUESTIONS

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Make every RPKI change explainable — from authorization to path to reachability.